Indian Institute of Technology Gandhinagar

Jan 9th Class is in AB1/103

Course No.: CS 303 **Course Title:** Mathematical Foundations for AI

Credits (L T P C): 3 1 0 4

Prerequisites (if any): None

Expected frequency of offering: Every year

Google Classroom Link: https://classroom.google.com/c/ODM4MjI5Nzc0OTQ0?cjc=bgp2udta

Course contents

Background and Numerical Foundations

Data representation & numerics: IEEE floating-point, rounding; under/overflow; catastrophic cancellation; conditioning vs ill-posedness; quick norms recap; why A^TA worsens conditioning.

Linear Algebra and Spectral Methods

Vector/row/column/null spaces; rank–nullity; orthogonality; projections; Least-Squares (LS) as projection; normal equations vs QR; pseudoinverse and minimum-norm LS. Eigen-decomposition (symmetric); SVD; Cholesky Decomposition; covariance matrices; PCA as SVD.

Multivariate Calculus and Differentiation

Gradients, Jacobians, Hessians; chain rule in vector/matrix form; trace-trick identities; finite-difference methods and pitfalls. Automatic differentiation (concept): computational graphs; gradient checking; where AD helps.

Probability and Estimation

Continuous multivariate probability: multivariate Gaussian; covariance/conditioning; marginal vs conditional; change of variables (brief). Estimation: likelihood; MLE vs MAP; conjugate priors (Beta–Binomial, Gaussian–Gaussian) with full worked examples.

Optimization Methods

Optimization—Taylor view & Gradient Descent (GD): Taylor link to methods; gradient descent; step-size choices; stopping criteria. Convexity: convex sets; convex and strongly convex functions. Second-order & Adaptive Methods: Newton intuition; quasi-Newton snapshot; Gauss–Newton and when it shines (LS problems); adaptive methods (Adam, RMSProp) intuition.

Constrained Optimization and Programming

Constrained optimization: Lagrangian; KKT conditions; primal/dual perspective; feasibility vs optimality; simple projected methods. Optimization Programs (Quadratic Programming (QP)/Least-Squares (LS) Focus): QP canonical form and geometry; QP ↔ LS link; brief note on LP/solvers.

Information Theory and Sequential Models

Information theory I: self-information, entropy; KL divergence; cross-entropy as Negative Log Likelihood. Information theory II: compression intuition; Huffman coding worked mini-example; "perfect communication over a noisy channel". Markov chains I: sequential modeling; joint probability of sequences; chain representation; parameter learning from complete data; smoothing/regularization notes. HMM (end): model setup; forward–backward and Viterbi at an applied-math level.

Learning outcomes

1. Diagnose numerical issues in AI computations and choose stable linear-algebraic formulations.
2. Derive and verify gradients/Jacobians/Hessians for ML objectives (analytic + finite-difference checks; AD conceptually).
3. Choose and justify optimization methods (GD, Gauss–Newton, constrained/KKT, LP/QP) with convexity in view.
4. Connect entropy/cross-entropy/KL to ML losses and coding arguments; execute a small Huffman example.
5. Model sequence data with Markov chains and carry out basic parameter learning; apply HMM inference on toy problems.

Grading policy

- **Problem Sets / Assignments / Tutorials:** 25% — The problem sets will be given to solve. Delayed submission will have penalties. Weightage will be 25%.
- **Tests (3 total):** 70% — There will be three tests including mid semester and end semester examination. Mid semester and end semester will have a weightage of 25% each. Remaining test will have a weightage of 20%.
- Attendance: Attendance will have a weightage of 5%.

Make-up tests will not be offered except under exceptional circumstances with valid documentation, and late submissions of assignments may be subject to penalties.

Texts/References

- **Deisenroth, M. P.; Faisal, A. A.; Ong, C. S.** Mathematics for Machine Learning.
- **Boyd, S.; Vandenberghe, L.** Convex Optimization.
- **MacKay, D.** Information Theory, Inference, and Learning Algorithms (entropy/KL/coding/Huffman).
- **Strang, G.** Linear Algebra and Learning from Data.
- **James, G., Witten, D., Hastie, T., Tibshirani, R., & Taylor, J.** An Introduction to Statistical Learning: With Applications in Python.
- **Murphy, K. P.** Probabilistic Machine Learning: An Introduction.

Tentative weekly plan

Week	Topics
1	Background & numerics: IEEE FP, rounding; under/overflow; catastrophic cancellation; conditioning vs ill-posedness; quick norms recap; why A^TA worsens conditioning.
2	Linear Algebra I: vector/row/column/null spaces; rank–nullity; orthogonality; projections; LS as projection; normal equations vs QR; pseudoinverse and minimum-norm LS.
3	Linear Algebra II (spectral): eigen-decomposition (symmetric); SVD; Cholesky Decomposition; covariance matrices; PCA as SVD.
4	Multivariate calculus I: gradients, Jacobians, Hessians; chain rule in vector/matrix form; trace-trick identities; finite-difference methods and pitfalls.
5	Differentiation II: automatic differentiation (concept), computational graphs,

	gradient checking; where AD helps.
6	Probability & estimation I: multivariate Gaussian; covariance/conditioning; marginal vs conditional; brief change of variables.
7	MID-SEM BREAK
8	MID-SEM BREAK .
9	Estimation II: likelihood, MLE vs MAP; conjugate priors (Beta–Binomial, Gaussian–Gaussian) with full worked examples.
10	Optimization I: Taylor view & Gradient Descent; step-size choices; stopping criteria.
11	Convexity: convex sets, convex & strongly convex functions.
12	Second-order & adaptive methods: Newton intuition; quasi-Newton snapshot; Gauss–Newton for LS; adaptive methods (Adam, RMSProp) intuition; practical pros/cons.
13	Constrained optimization: Lagrangian; KKT; primal/dual perspective; feasibility vs optimality; simple projected methods.
14	Optimization programs: QP canonical form & geometry; QP ↔ LS link; brief note on LP/solvers.
15	Information theory I: self-information, entropy; KL divergence; cross-entropy as Negative Log Likelihood.
16	Information theory II and sequential models: Huffman demo; noisy-channel compression intuition; Markov chains; HMM (forward–backward, Viterbi) overview.

Any other remarks

Nil.